

SILSTRUP, Denmark

WMO: 060190

Lat: 56.9300N

Lon: 8.6412E

Elev: 41

StdP: 100.83

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.9	-5.2	-9.5	1.7	-5.2	-8.0	1.9	-4.0	25.5	6.0	22.3	5.2	4.0	60	0.755

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	24.5	18.3	22.6	17.5	21.0	16.7	19.1	22.9	18.2	21.5	17.4	20.1	4.6	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	12.8	20.4	16.9	12.1	19.5	16.1	11.5	18.7	54.5	23.2	51.7	21.8	49.0	20.2	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.3	17.9	14.6	DB	-7.8	27.7	2.5	2.2	-9.6	29.3	-11.1	30.6	-12.5	31.8	-14.3	33.4	(4)
(5)			WB	-8.3	20.6	2.5	1.3	-10.1	21.6	-11.6	22.4	-13.0	23.1	-14.8	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	1.4	1.4	3.4	7.2	10.8	13.7	16.4	16.2	13.8	9.9	6.1	3.6
	DBStd	6.08	3.49	3.41	3.15	2.79	3.14	2.57	2.66	2.02	2.20	2.48	2.96	3.69
	HDD10.0	1180	266	242	204	92	27	1	0	0	1	31	118	199
	HDD18.3	3540	524	475	462	333	235	141	76	72	139	261	366	458
	CDD10.0	706	0	0	0	9	51	112	197	193	113	29	2	0
	CDD18.3	25	0	0	0	0	1	3	14	6	1	0	0	0
	CDH23.3	110	0	0	0	0	1	14	71	22	3	0	0	0
	CDH26.7	12	0	0	0	0	0	1	8	3	0	0	0	0
Wind	WSAvg	6.4	7.3	7.0	6.8	5.8	6.1	6.0	5.5	5.7	6.0	6.5	6.8	7.3
Precipitation	PrecAvg	820	75	54	46	42	47	59	61	80	88	101	87	80
	PrecMax	1078	120	123	95	95	98	144	102	154	192	196	182	167
	PrecMin	539	9	15	4	9	3	1	9	19	31	31	25	23
	PrecStd	137	31	29	24	22	23	33	28	35	42	44	40	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	8.3	12.1	18.6	22.8	25.0	27.9	25.8	23.2	17.4	13.4	10.7
		MCWB	8.4	7.2	8.8	13.0	16.3	18.1	19.0	18.9	17.7	15.0	12.6	9.9
	2%	DB	8.1	7.2	9.4	15.8	20.5	22.7	25.2	23.1	20.1	15.5	11.9	9.7
		MCWB	7.4	6.3	7.4	11.0	15.3	16.8	18.7	17.6	16.3	14.0	11.2	8.9
	5%	DB	7.2	6.5	8.0	13.5	18.3	20.3	23.3	21.2	18.3	14.4	11.0	8.9
		MCWB	6.5	5.8	6.6	10.2	14.0	16.1	18.4	16.8	15.5	13.1	10.3	8.2
	10%	DB	6.4	5.8	7.1	11.8	16.1	18.2	21.2	19.8	17.1	13.4	10.2	8.1
		MCWB	5.7	5.1	6.0	9.0	12.8	14.8	17.3	16.3	14.8	12.3	9.5	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	7.5	9.0	13.4	17.5	19.1	20.6	20.0	18.4	15.6	12.7	10.2
		MCDB	9.0	7.8	11.3	16.8	21.2	23.6	24.7	23.1	20.9	16.8	13.1	10.7
	2%	WB	7.6	6.6	7.8	11.6	16.1	17.7	19.4	18.7	17.0	14.4	11.2	9.2
		MCDB	8.0	7.0	8.9	14.6	19.3	21.6	23.5	21.7	19.7	15.2	11.6	9.5
	5%	WB	6.6	5.9	6.9	10.4	14.5	16.5	18.6	17.7	16.1	13.4	10.5	8.4
		MCDB	7.1	6.4	7.8	13.0	17.5	19.6	22.4	20.0	17.9	14.1	11.0	8.8
	10%	WB	5.8	5.3	6.2	9.2	13.1	15.2	17.7	16.9	15.3	12.5	9.5	7.6
		MCDB	6.4	5.8	6.9	11.4	15.5	17.6	20.9	18.9	16.7	13.2	10.1	8.1
Mean Daily Temperature Range	5% DB	MDBR	3.3	3.3	4.3	6.0	6.5	6.4	6.8	6.3	5.1	4.1	3.3	3.2
		MCDBR	3.8	3.9	5.2	9.0	9.7	9.8	9.7	8.5	7.0	4.8	3.7	3.3
		MCWBR	4.0	3.5	3.7	5.4	5.8	5.6	4.7	4.2	4.1	4.0	3.7	3.6
	5% WB	MCDBR	3.7	3.8	4.9	8.3	8.9	9.0	8.9	7.3	6.4	4.7	3.5	3.4
		MCWBR	4.0	3.7	3.8	5.4	5.8	5.7	4.8	4.3	4.4	4.1	3.7	3.8
Clear-Sky Solar Irradiance	taub	0.314	0.336	0.353	0.388	0.378	0.370	0.386	0.389	0.368	0.348	0.324	0.298	
	taud	2.217	2.285	2.336	2.290	2.383	2.428	2.401	2.396	2.439	2.405	2.287	2.198	
	Ebn at Noon	547	706	801	824	860	871	847	817	781	690	534	457	
	Edh at Noon	54	76	93	113	109	106	107	101	84	67	50	42	
All-Sky Solar Radiation	RadAvg	0.41	0.97	2.36	4.00	5.31	5.80	5.32	4.18	2.81	1.39	0.52	0.29	
	RadStd	0.08	0.16	0.21	0.47	0.50	0.46	0.54	0.31	0.29	0.17	0.09	0.03	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	-127	-169	N/A	N/A

Nomenclature: See separate page